Introduction to AI Safety, Ethics, and Society

Normative Ethics

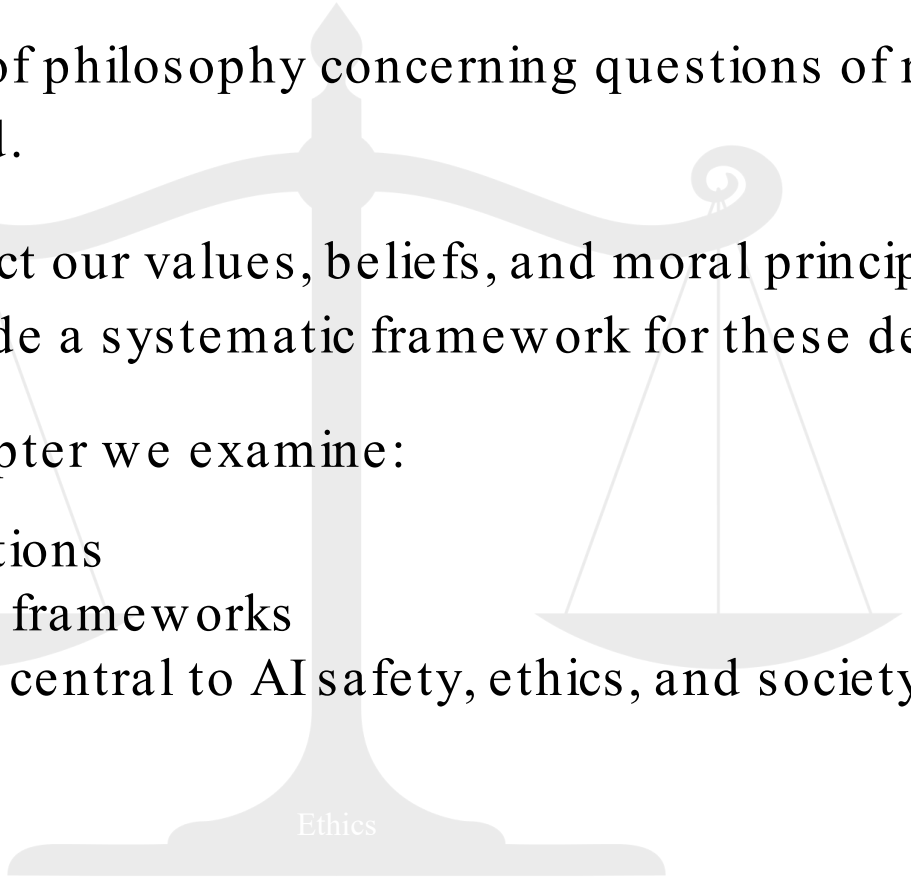
Introduction

Ethics is the branch of philosophy concerning questions of right and wrong, good and bad.

Moral decisions reflect our values, beliefs, and moral principles. Ethics seeks to provide a systematic framework for these decisions.

Throughout this chapter we examine:

1. Moral considerations
2. Common ethical frameworks
3. Ethical concepts central to AI safety, ethics, and society



Roadmap

1. Why learn about ethics?

1. Normative factors: wellbeing, constraints, impartiality, etc.

1. Moral Theories:
    - Utilitarianism
    - Deontology

Why Learn About Ethics?

To develop a foundation for understanding AI safety discourse. This is important because…

1. Widespread use and development of AI systems is taking place throughout various domains of human life.

2. Ensuring that AI systems are aligned with human values reduces the chance of bad outcomes.

3. AI systems generate ethical questions that are unique to their technological nature and capabilities.

Ethics and Relativism

It may be that no moral principles or judgements are absolutely or universally correct; instead, they are only correct within a culture.

Moral Relativism is the view that there is no objective standard of morality that applies to all people across all situations.



Problems with moral relativism:

1. Extremely harmful actions are not permissible just because they are culturally relative.
2. There may exist moral disagreements within cultures.

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Decomposing Goodness

Moral decisions often consider the intrinsic and instrumental goods that are at stake.

Instrumental goods are valuable because of the benefits or outcomes they provide – for example, money, power, career opportunities.

Intrinsic goods are valuable in and of themselves; common supposed examples are love, pleasure, beauty, and truth.

Some philosophers think there are many intrinsic goods, others accept none, or only one — wellbeing.

Wellbeing

How wellbeing is defined is debatable; however, it is commonly considered to be an **intrinsic good**.

Wellbeing generally refers to how well a person's life is going for them, or whether a person is happy, healthy, and fulfilled.

Throughout this section, we discuss three common accounts of wellbeing:

1. Wellbeing as pleasure.
2. Wellbeing as preference satisfaction.
3. Wellbeing as a set of objective goods.



Wellbeing as Pleasure

Wellbeing as pleasure: the balance of pleasure (or happiness) over pain (or suffering) is what constitutes wellbeing. Also called *hedonism*.

While we may all have different preferences and desires, pleasure seems to be universally valued. All other goods might be instrumental.

However, some philosophers argue against this view because it does not consider other ostensibly relevant factors to our wellbeing such as the pursuit of knowledge.

Wellbeing as Preference Satisfaction (1/2)

Wellbeing as preference satisfaction: what really matters for wellbeing is that our desires or preferences are satisfied.

Different types of preferences might differ greatly in different situations. Consider Alice's preferences over different candidates in an election:

1. **Stated preference** → "I prefer AI's output X"
2. **Revealed preference** → I chose AI's output X
3. **Informed preference** → I would have chosen AI's output X if I knew …

Which preferences matter?

Wellbeing as Preference Satisfaction (2/2)

Stated preferences are outwardly expressed, but sometimes, they conflict with revealed preferences. A **revealed preference** illustrates our actions, not our intentions and they can be more useful than stated preferences.

> A voter may express a preference for Google over ChatGPT, but when they actually have queries, they may more frequently use ChatGPT.

An **informed preference** considers all relevant information and can change as more information is acquired. These help us to uncover the reasoning behind stated/revealed preferences to better understand how they might change or conflict.

Wellbeing as Objective Goods

Objective goods are those things which are good independent of personal beliefs or opinions – they are *universal*.

An example set: moral goodness, rational activity, development of abilities, having children and being a good parent, knowledge, awareness of true beauty.



We may not always agree on what constitutes an objective good, and whether it can be considered universal.

Wellbeing and AI

AI influence on human wellbeing: AI chatbot companions could influence our wellbeing

- If they aim to maximize revealed preferences, they may addict us and leave us unhappy
- If they aim to maximize pleasure, they may give us amusing but shallow content
- If they aim to maximize objective goods, they may push us to pursue self-development goals that we may not wish

Wellbeing of AIs: if AIs are conscious, then

- Many coherently-acting AIs could have wellbeing according to a preference view of wellbeing
- AIs that can reason, gain knowledge, or develop their abilities could have wellbeing according to an objective goods view

Wellbeing Summary

In this section, we have covered the following accounts of wellbeing:

1. Wellbeing as pleasure
2. Wellbeing as preference satisfaction
3. Wellbeing as a set of objective goods

Though these views differ from one another they all share a **common goal**: promoting actions that result in morally good outcomes.

A morally good outcome will almost always aim to affect wellbeing to some degree.

Obligations, Constraints and Rights (1/2)

Constraints are actions that we're morally prohibited from taking; common examples are killing or lying.

Obligations are actions that we're morally required to take; common examples are keeping our promises or telling the truth.

Rights are claims that individuals may have over their community.

- Rights that require certain actions from others are called "**positive rights**" → access to education, food, or shelter.
- Rights that require others to abstain from certain behaviors are called "**negative rights**" → censorship or discrimination.

Some say AIs should eventually get negative rights but not positive rights.

Partiality and Impartiality (1/2)

Partial moral theories imply that we should value individual lives preferentially. These emphasize **special obligations**.

- We have special obligations towards ourselves, our friends, our communities, or our families that we do not have toward strangers.

Impartial moral theories imply that we should value individual lives equally. These disregard special obligations.

Most modern moral theories are impartial, and almost every moral theory requires impartiality in at least some contexts.

Partiality and Impartiality (2/2)

Most modern theories agree that all human beings, regardless of their differences, are included in the *moral circle* – they are subjects of moral concern.

Some philosophers, such as hedonists, have also argued that non-human animals should be included, due to a capacity for suffering.

But perhaps we should extend the moral circle even further, considering entities that don't have the same kinds of conscious experience as current humans, such as future humans or AIs.

Impartial AIs care for humans initially but come to care for them less.

Recap

Understanding ethics is important because AI systems—likely to have serious impacts on the world—must be used ethically. In particular, we must ensure that AI systems are aligned with human values.

Three conceptions of wellbeing suggest different ways to use AIs.

Moral considerations like impartiality, obligations, constraints, and rights give us a shared language to describe and debate the ethics of AIs and their impacts. These considerations are all relevant to common-sense morality and the following moral theories.

Roadmap

1. Why learn about ethics?

1. Moral considerations: wellbeing, obligations, partiality, et cetera.

1. Moral Theories:
    - Utilitarianism
    - Deontology

From Considerations to Theories

Moral considerations are part of **metaethics**: the underlying concepts and assumptions that make moral reasoning possible.

Normative ethics is about concrete moral standards and principles that govern how we ought to behave. Here, we consider two of the most common theories in academic ethics:

1. Utilitarianism
2. Kantian Ethics/Deontology

Utilitarianism Overview

Utilitarianism seeks to maximize overall wellbeing → the right action is the one that increases overall wellbeing the most.

According to utilitarianism, the right action in any situation is the one which will increase overall wellbeing the most—not just for the people directly involved in the situation, but globally.

When we maximize wellbeing, we can make moral questions into empirical ones.

Utilitarianism: Drunk Driving (1/2)

Drunk driving. Amanda has had a few drinks, and is deciding whether to drive or take the bus home. What should she do?

Utilitarians would analyze the situation by…

1. Listing all the possible outcomes of driving vs. not driving.
2. Calculating the probabilities of each outcome.
3. Calculating the '**utility**' – wellbeing – brought about by each outcome.

Expected utility of each choice is the sum of the probabilities of each outcome multiplied by its utility.

Utilitarianism: Drunk Driving (2/2)

Amanda's action	Possible outcome(s)	Probability of each outcome	Utility
Amanda takes the bus.	Amanda is frustrated, the bus is slow, and she has to wait in the cold.	1	-1
Amanda drives home.	Amanda gets home safely, far sooner than she would have on the bus.	.95	+1
	Amanda gets into an accident and someone is fatally injured.	.05	-1000

*the probability and utility values we assign here are arbitrary, and only serve to illustrate what such a calculation might look like.

Expected utility:

Amanda gets the bus → 1 x - 1 = **- 1**

Amanda drives home →

0.95 x 1 + (0.05 x - 1000) = - 49.05

Amanda should take the bus!

Utilitarianism's Claims

Utilitarianism is a form of consequentialism → *consequences determine whether an action is good or bad.*

We measure* the effects of actions on wellbeing alone → *wellbeing is the only intrinsic good.*

All people have the same intrinsic moral worth → *everyone's wellbeing should be weighed impartially.*

It is insufficient to do good, we must do what is best → *we should maximize wellbeing.*

**We measure wellbeing according to our choice of theory, e.g. Hedonism vs. Preference Satisfaction.*

Critiques of Utilitarianism (1/3)

Utilitarianism is too demanding → If our money is more helpful to others than it is to us, there isn't a utilitarian reason to keep it. For instance, **Peter Singer** has argued that we should donate at least a third of our income to charity.

Defences include:

1. **Utilitarianism is not as demanding in practice as it is in theory.** Often, acting in a utilitarian way is quite "normal", avoiding issues like burnout.
2. **We live in a demanding world.** Maybe most of us should donate more.

Utilitarianism over-emphasizes wellbeing → **Robert Nozick** famously highlights this point in his popular thought experiment.

Critiques of Utilitarianism (2/3)



Experience Machine: *Consider an experience machine offering any desired sensation. Neuropsychologists could make you believe you're achieving great feats while you float in a tank, your brain connected to electrodes. Would you choose to live connected to this machine, pre-set with life's experiences?* **Most people say no.**

What if we reverse the thought experiment?

You 'wake up' in an Experience Machine tomorrow. You can either forget this, returning to your current (simulated) life or return to your "real" one.

Most people choose to stay in the machine. This is *status quo bias*.

Critiques of Utilitarianism (3/3)

Utilitarianism requires intractable reasoning → we cannot reliably and precisely calculate future utility values. It is also impractical to expect this. The drunk driving example is idealized; for instance, we did not list all possible outcomes or consider long-term consequences.

The utilitarian response: *a theory's criterion of rightness doesn't need to take the same form as its decision procedure*.

> A **criterion of rightness** → whatever a theory claims makes an action right.
>
> A **decision procedure** → the process a theory recommends that individuals use to make decisions.

Deontology

Deontology's key ideas:

1. Focus on *constraints and obligations*, not the best outcome.
2. Formulation of rules or principles that protect *human autonomy*.
3. The *rights and intentions* of individuals are morally relevant.

Deontological rules from the Ten Commandments: "*Thou shalt not kill*", "*Thou shalt not steal*", "*Honor thy mother and father*", etc.

Deontological theories are systems of moral rules, rights, duties and obligations which constrain our behaviour. These theories typically present themselves as improvements over consequentialism.

Deontology's Critique of Utilitarianism

Consequentialism doesn't allow options → important life decisions like marriage or career choice will have "correct" utility-maximizing answers.

> Deontologists say this view is too demanding – to preserve human autonomy, we provide a list of impermissible actions, and allow choice to govern the rest.

Consequentialism doesn't forbid extreme actions → killing innocents, torturing – these actions aren't *always* wrong for consequentialists.

> Deontologists argue some actions are simply forbidden by appeal to universal constraints or obligations.

Deontology: The Doctrine of Double Effect

Deontology also emphasizes the **intentions** of agents in their moral considerations. Consider **The Doctrine of Double Effect**:

> An agent is morally allowed to carry out actions that predictably lead to bad outcomes as long as they intend the good effect, but not the bad effect of the action.

Siamese twins: a pair of siamese twins needs a medical procedure, without which both of them would die. However, one of them will die as a consequence. The doctrine of double effect says the procedure is justified insofar as we *intend* to save one of the twins.

Deontology: Act/Omission

Intuitively, we do not hold a person responsible for what they did not do. In deontology, this is the **Act vs. Omission** distinction.

> ***Stealing***: While walking past a bank at night, Bob notices the night deposit box is open. Inside it, he sees a bag, filled with money. He decides to steal the bag. Alice sees Bob taking the money, but never bothers to report it.



Deontologists would not hold Alice responsible for Bob's actions.

Critiques of Deontology

Deontology responds unconvincingly to moral catastrophes.

> ***Nuclear Terrorism***. The only way to prevent a nuclear terror attack is by torturing the perpetrator. But torture is impermissible to deontologists, regardless of the consequences.

In response, some deontologists have adopted a **threshold**: when enough lives are at stake, the theory defers to consequentialism. However, there is no clear, non-arbitrary way to determine "enough".

Deontology holds constraints like promises over vastly better outcomes.

Chapter Recap

Ethics is the study of moral principles that guide decisions and actions.

AI researchers can use ethics to guide development and governance.

Moral considerations include goodness, constraints, special obligations, and more.

Utilitarianism focuses on consequences and wellbeing; deontology on constraints.